Introduction to Stata

Lecture 4: Linear regression (OLS)

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Where we are, where we go

From Lecture 3 (`data/clean/`)

- `shiw_2022.dta`: households, 2022, with `sample`, `y_eq_w`, `ln_y_eq_w`
- `individuals_full.dta`: persons, with personal income

Missing a file? Run `L2_solutions.do` and `L3_solutions.do`.

Handbook: Chapter 8; § 4.5 (factor variables).

Today

$$\ln y_i = \beta_0 + \beta_1 s_i + \beta_2 a_i + \beta_3 a_i^2 + \cdots + u_i$$

s: years of schooling, a: age of the main earner

Estimate, read, interpret; robust standard errors; predictions; tests; diagnostics; a table for a paper.

PART 1

Demonstration

regress and its output

Handbook § 8.1, 8.2

```
use "$clean/shiw_2022.dta", clear
keep if sample == 1
regress ln_y_eq head_educ_years
```

Number of obs	9,617
$\hat{\beta}_{\text{school}}$ (se)	0.083 (0.0014)
R-squared	0.274

Read, in this order

1. `Number of obs`: compare with the data
2. The coefficient: sign, size, units
3. Std. err., t, p-value, 95% CI
4. R^2: share of the variance of ***y*** explained
5. F test: all slopes jointly zero

Significant $\neq$ large; not significant $\neq$ zero.

From Lecture 3: slope and correlation

Handbook § 8.8

```
scalar b_ols = _b[head_educ_years]          // coefficient
scalar r2_ols = e(r2)                       // stored results in e()
quietly correlate ln_y_eq head_educ_years
display "OLS slope: " b_ols "   rho^2: " r(rho)^2 "   R2: " r2_ols
```

- Simple regression: $\hat{\beta}_1 = \hat{\rho}\,\hat{\sigma}_y/\hat{\sigma}_x$ and $R^2 = \hat{\rho}^2 = 0.52^2 = 0.27$
- `_b[x]`, `_se[x]`: coefficient and standard error; `e(N)`, `e(r2)`, `e(sample)`

Interpreting the coefficients

Handbook § 8.3

Model	Specification	$\Delta x = 1$ (or 1%) $\Rightarrow$
level-level	$y = \alpha + \beta x$	$\Delta y = \beta$ units
log-level	$\ln y = \alpha + \beta x$	$\Delta y \approx 100\beta$ % (exact: $100(e^{\beta} - 1)$)
level-log	$y = \alpha + \beta \ln x$	$\Delta y \approx \beta/100$ units
log-log	$\ln y = \alpha + \beta \ln x$	$\Delta y \approx \beta$ % (elasticity)

Our case (log-level): one more year of schooling of the main earner → about 8.3% higher equivalised income (exact: 8.7%).

This is an *association*: schooling is correlated with ability, family background, sector… Causality needs a research design (IV, Chapter 10).

Multiple regression and factor variables

Handbook § 4.5, 8.3

```
regress ln_y_eq head_educ_years c.head_age##c.head_age hhsize ///
        i.head_female i.area3
```

- `c.` continuous, `i.` categorical (base omitted), `##` all interaction terms
- `c.age##c.age` = age and age^2
- Each coefficient: holding the others constant

Schooling	0.088
Female main earner	−0.179
Centre (vs North)	−0.135
South and Islands (vs North)	−0.392
R^2	0.40

The schooling coefficient rises (0.083 → 0.088): older main earners have less schooling and higher income, so omitting age biased it down. Female: $100(e^{-0.179} - 1) = -16.4\%$.

Standard errors

Handbook § 8.4

```
regress ln_y_eq head_educ_years c.head_age##c.head_age hhsize ///
        i.head_female i.area3, vce(robust)
```

- Default: errors homoskedastic. `vce(robust)`: heteroskedasticity-robust (HC1)
- Coefficients are identical; only standard errors, t, p-values and CIs change
- In cross-sections, robust standard errors are the standard choice
- `vce(cluster ireg)`: correlation within regions allowed, but only 20 clusters: unreliable (fewer than about 30–50)

Predictions and residuals

Handbook § 8.5

```
predict ln_y_hat if e(sample), xb          // fitted values
predict u_hat    if e(sample), residuals   // residuals
summarize ln_y_eq ln_y_hat u_hat           // mean of residuals = 0
```

- `if e(sample)`: only the observations used in the estimation
- After a regression of $\ln y$, `predict` gives $\widehat{\ln y}$; $\exp(\widehat{\ln y})$ underestimates $E[y \mid x]$

Tests and linear combinations

Handbook § 8.6

```
testparm i.area3              // H0: no difference across areas (joint)
test 2.area3 = 3.area3        // H0: Centre = South
lincom 3.area3 - 2.area3      // size of South - Centre, with SE and CI
nlcom -_b[head_age] / (2*_b[c.head_age#c.head_age])   // nonlinear
```

- `test`: F (Wald) test of one or several linear restrictions
- `lincom`: the size of a combination, not only a test
- A joint test is not the same as separate t-tests: two correlated regressors can be individually not significant but jointly significant

A table for a paper

Handbook § 8.9

```
quietly regress ln_y_eq head_educ_years, vce(robust)
estimates store m1
quietly regress ln_y_eq head_educ_years c.head_age##c.head_age hhsize, vce(robust)
estimates store m2
...
etable, estimates(m1 m2 m3) column(estimates) mstat(N) mstat(r2) ///
        showstars showstarsnote
collect style cell result[_r_b _r_se], nformat(%9.4f)
collect export "$tables/ols_households.tex", replace
```

`etable` is built in (Stata 17+). Alternative: `esttab` from the **`estout`** package (`ssc install estout`), § 8.9.

Commands of the day

`regress y x1 x2`	OLS	`test`, `testparm`	Wald tests
`i.`, `c.`, `#`, `##`	factor variables	`lincom`, `nlcom`	combinations
`vce(robust)`	robust SEs	`estat hettest/ovtest/vif`	diagnostics
`predict, xb / residuals`	fitted, residuals	`rvfplot`	residual plot
`_b[]`, `_se[]`, `e()`	stored results	`estimates store/table`	keep models
`[pweight = peso]`	population	`etable`, `collect export`	tables

Beyond this introductory course: binary outcomes (Ch. 9), instrumental variables (Ch. 10), panel data (Ch. 11) with the same SHIW panel households.

PART 2

Now let's practice on Stata

Open `L4_exercises.do`, Part 2: five exercises, 10 minutes each, then we write the solution together.